

Rockchip RK3566_RK3568 Linux6.1 SDK

Release Note

ID: RK-RB-YF-961

Release Version: V1.0.1

Release Date: 2024-09-20

Security Level: ☐Top-Secret ☐Secret ☐Internal ☒Public

DISCLAIMER

THIS DOCUMENT IS PROVIDED "AS IS". ROCKCHIP ELECTRONICS CO., LTD. ("ROCKCHIP") DOES NOT PROVIDE ANY WARRANTY OF ANY KIND, EXPRESSED, IMPLIED OR OTHERWISE, WITH RESPECT TO THE ACCURACY, RELIABILITY, COMPLETENESS, MERCHANTABILITY, FITNESS FOR ANY PARTICULAR PURPOSE OR NON-INFRINGEMENT OF ANY REPRESENTATION, INFORMATION AND CONTENT IN THIS DOCUMENT. THIS DOCUMENT IS FOR REFERENCE ONLY. THIS DOCUMENT MAY BE UPDATED OR CHANGED WITHOUT ANY NOTICE AT ANY TIME DUE TO THE UPGRADES OF THE PRODUCT OR ANY OTHER REASONS.

Trademark Statement

"Rockchip", "瑞芯微", "瑞芯" shall be Rockchip's registered trademarks and owned by Rockchip. All the other trademarks or registered trademarks mentioned in this document shall be owned by their respective owners.

All rights reserved. ©2024. Rockchip Electronics Co., Ltd.

Beyond the scope of fair use, neither any entity nor individual shall extract, copy, or distribute this document in any form in whole or in part without the written approval of Rockchip.

Rockchip Electronics Co., Ltd.

No.18 Building, A District, No.89, software Boulevard Fuzhou, Fujian, PRC

Website: www.rock-chips.com

Customer service Tel: +86-4007-700-590

Customer service Fax: +86-591-83951833

Customer service e-Mail: fae@rock-chips.com

Preface

Overview

This document primarily introduces the Rockchip RK3566_RK3568 Linux 6.1 SDK release notes, aiming to assist engineers in quickly mastering the development of the RK3566_RK3568 Linux 6.1 general SDK and related debugging methods.

Target Audience

This document (this guide) is mainly applicable to the following engineers:

Technical Support Engineers

Software Development Engineers

Chip System Support Status

Chip Name	Buildroot	Debian	Yocto
RK3566	Y	Y	Y
RK3568	Y	Y	Y

Revision Record

Date	Version	Author	Modification Description
2024-06-20	V1.0.0	Caesar Wang	Initial version.
2024-09-20	V1.0.1	Caesar Wang	Fixes typo. Match SDK Version

Table of Contents

Rockchip RK3566_RK3568 Linux6.1 SDK Release Note

1. Overview
2. Main Supported Features
 - 2.1 Hardware Capabilities
3. SDK Acquisition Instructions
 - 3.1 Downloading via Code Server
 - 3.2 Obtaining SDK Source Code through Local Decompression
4. Software Development Guide
5. Hardware Development Guide
6. SSH Public Key Operation Instructions
 - 6.1 Key Permission Management
 - 6.2 Reference Documentation

1. Overview

This SDK supports three systems, based on Buildroot 2024.02, Debian 12, and Yocto 5.0, with the kernel based on Kernel 6.1 and booting based on U-boot v2017.09. It is suitable for RK3566/RK3568 EVB development boards and all Linux products that are developed based on these development boards.

The development package is applicable to, but not limited to, cloud terminals/industrial motherboards and other AIoT products, providing a flexible data path combination interface to meet customers' customized needs for free combination. For specific function debugging and interface descriptions, please read the documents in the docs/ directory of the project.

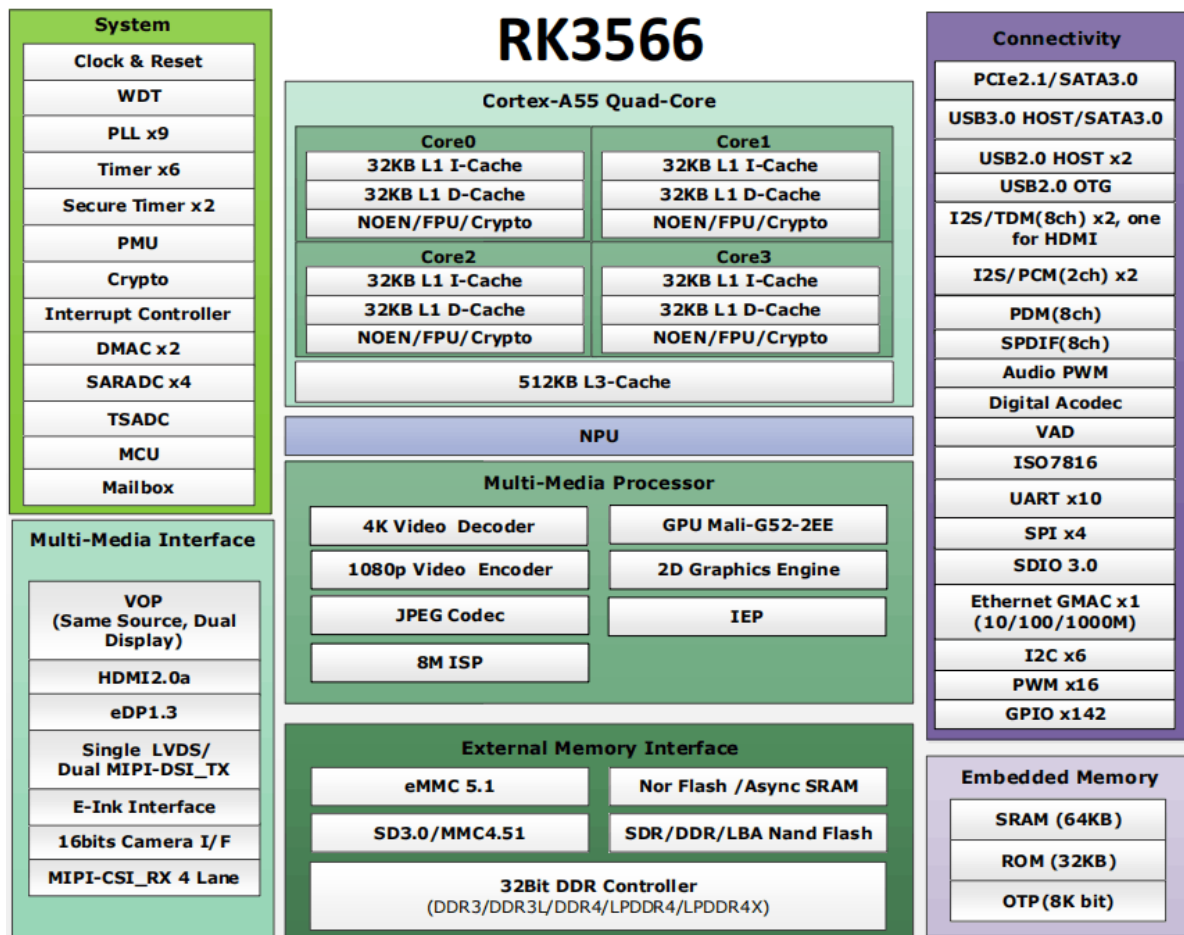
2. Main Supported Features

Feature	ModuleName
System	Debian, Yocto, Buildroot
Partition Table	uboot, misc, boot, recovery, rootfs, oem, userdata
Filesystem Types	EXT2/3/4, VFAT, NTFS, UBIFS, SquashFS
Upgrade Recovery	OTA, AB, Recovery
Secure Boot	SecureBoot
Stress Test Tools	ROCKCHIP_TEST
Data Communication	Wi-Fi, Ethernet Card, USB, SD Card, SATA, PCI-e Interface
Applications	Multimedia Playback, Camera Browsing, Desktop UI, Browser

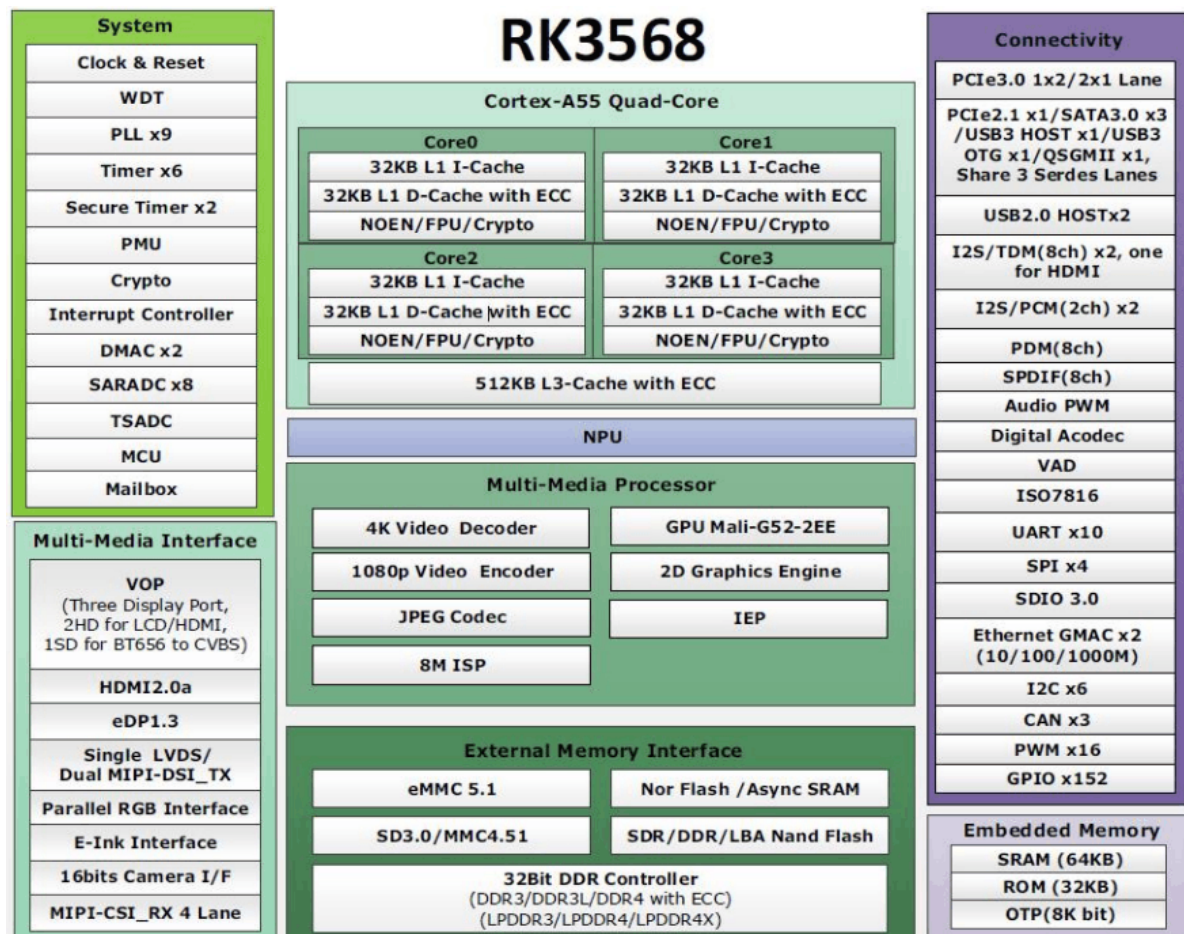
2.1 Hardware Capabilities

Specific interface functions of the hardware,

- For RK3566 chips, refer to the following chip diagram



- For RK3568 chips, refer to the following chip diagram



3. SDK Acquisition Instructions

The SDK is released and obtained through the Rockchip code server. For its compilation and development environment, refer to Chapter 4 [Software Development Guide](#).

3.1 Downloading via Code Server

To obtain the RK3566/RK3568 Linux software package, access to the source code repository provided by Rockchip is required through an account. Customers apply for the SDK from the Rockchip technical support window, and simultaneously provide an SSH public key for server authentication and authorization. Once authorization is granted, synchronization of the code is possible. For information on SSH public key authorization for Rockchip's code server, please refer to Section 6 [SSH Public Key Operation Instructions](#).

The command for downloading the RK3566/RK3568 Linux SDK is as follows:

```
repo init --repo-url https://gerrit.rock-chips.com:8443/repo-release/tools/repo -
u https://gerrit.rock-chips.com:8443/linux/rockchip/platform/manifests -b
rk3566_rk3568 -m rk3566_rk3568_linux6.1_release.xml
```

The `repo` is a Python script written by Google to invoke git, primarily used for downloading and managing the software repository of projects. The download address is as follows:

```
git clone https://gerrit.rock-chips.com:8443/repo-release/tools/repo
```

3.2 Obtaining SDK Source Code through Local Decompression

For the convenience of customers to quickly obtain SDK source code, Rockchip's technical support usually provides the initial compressed package of the corresponding version of the SDK. Developers can obtain the initial compressed package of the SDK code through this method, and the source code obtained after decompression is consistent with the source code downloaded through repo.

Taking RK3566_RK3568_LINUX6.1_SDK_RELEASE_V1.0.0_20240620.tgz as an example, after copying to the initialization package, the source code can be checked out with the following command:

```
mkdir rk3566_rk3568
tar xvf RK3566_RK3568_LINUX6.1_SDK_RELEASE_V1.0.0_20240620.tgz -C rk3566_rk3568
cd rk3566_rk3568
.repo/repo sync -l
.repo/repo sync -c
```

Subsequently, developers can synchronize updates according to the update instructions regularly published by the FAE window using the `.repo/repo sync -c` command.

If there are issues with the repository download, the `--force-sync` parameter can be used to force an update, such as `.repo/repo sync -c --force-sync`. Before this, please ensure that local modifications have been backed up.

After updating the SDK code, a clean operation is required, for example: `./build.sh cleanall`

Explanation:

The software release version can be viewed through the project XML, and the specific method is as follows:

```
.repo/manifests$ realpath rk3566_rk3568_linux6.1_release.xml
For example: The printed version number is v1.0.0, and the update time is
20240620
<SDK>/repo/manifests/release/rk3566_rk3568_linux6.1_release_v1.0.0_20240620.xml
```

4. Software Development Guide

For software development related to the project, please refer to the Quick Start documentation under the project directory:

```
<SDK>/docs/en/RK3566_RK3568/Quick-start/Rockchip_RK356X_Quick_Start_Linux_EN.pdf
```

5. Hardware Development Guide

For hardware development related information, please refer to the user guide and hardware design guide documents in the project directory:

```
<SDK>/docs/en/RK3566_RK3568/Hardware/
├─ Rockchip_RK3566_EVB2_User_Guide_V1.1_EN.pdf
├─ Rockchip_RK3566_Hardware_Design_Guide_V1.1_20220206_EN.pdf
├─ Rockchip_RK3568_EVB_User_Guide_V1.2_EN.pdf
├─ Rockchip_RK3568_Hardware_Design_Guide_V1.2_20220126_EN.pdf
```

6. SSH Public Key Operation Instructions

Please follow the instructions in the "Rockchip_User_Guide_SDK_Application_And_Synchronization_EN" document to generate an SSH public key and send an email to fae@rock-chips.com to apply for access to the SDK code.

This document will be released for customer use during the process of applying for access rights.

6.1 Key Permission Management

The server can monitor the download count and IP information of a specific key in real time, and if any anomalies are detected, the download permission for the corresponding key will be disabled.

Please keep your private key file in a safe place and do not re-authorize it for use by a third party.

6.2 Reference Documentation

For more detailed instructions, please refer to the document at

`<SDK>/docs/en/Others/Rockchip_User_Guide_SDK_Application_And_Synchronization_EN.pdf`.